Stacked Waves and Qualitative Analysis of an Oncolytic Virus-Immune Model

Negar Mohammadnejad *, Thomas Hillen †

Dedicated to the memory of Dr. Mary Ann Horn

Abstract

Oncolytic virotherapy (OVT) represents a promising therapeutic method for cancer treatment. It involves the introduction of oncolytic viruses into the patient, which are engineered to selectively target and lyse tumor cells. Based on previous mathematical modeling of oncolytic viruses, we consider a mathematical model that describes the intricate interactions between the oncolytic virus, cancer cell populations, and the immune system. Our study includes a detailed qualitative and quantitative analysis of the model to explain why, despite their promise, oncolytic viruses alone rarely lead to complete and lasting regression of established tumors in vivo. We use parameter sensitivity analysis to support our findings. Furthermore, we consider the spatial version of the model in the form of a system of reaction-diffusion equations. A travelling wave analysis shows an unexpected phenomenon. The solution components do not necessarily evolve into a single traveling front; rather, they develop into stacked fronts, where each front propagates at a different speed. We give explicit formulas for these different invasion speeds, confirm those through numerical simulations, and discuss their significance for OVT.

Keywords: Mathematical oncology, oncolytic virotherapy, spreading speed, travelling Waves, stacked waves

1 Introduction

Oncolytic virotherapy (OVT) is an emerging cancer treatment in which oncolytic viruses are used to eradicate tumor cells. Oncolytic viruses (OVs) are able to selectively infect, replicate, and propagate within tumors (Engeland et al., 2020). The first genetically engineered OV, a herpes simplex virus 1 (HSV-1) mutant with deficient thymidine kinase, was developed in 1991 and used to treat malignant glioma in nude mice. These viruses selectively infect and destroy cancer cells, often engaging dendritic cells to enhance immune-mediated antitumor responses (Kim et al., 2015). Although oncolytic viruses (OVs) have advanced into clinical trials, with T-VEC receiving FDA approval for melanoma, significant

*Department of Mathematical and Statistical Sciences, University of Alberta, Edmonton, AB, Canada, T6G 2R3, negar3@ualberta.ca

†Department of Mathematical and Statistical Sciences, University of Alberta, Edmonton, AB, Canada, T6G 2R3, thillen@ualberta.ca (Corresponding author)

challenges remain, including limited viral infection efficiency and rapid immune clearance, which must be addressed to fully realize their therapeutic potential (Jafari et al., 2022).

The immune system plays a pivotal role in OV–cancer interactions. It was once considered as a virus replication inhibitor and a limiting factor in virus-cancer interactions. However, the interactions among cancer, virus and immune response are more complex. The immune system recognizes and attacks the tumor after the viral infection enhances the visibility of the tumor to the immune system. Moreover, the release of tumor antigens into the microenvironment, triggered by virus-induced lysis of cancer cells, helps reverse immune suppression, which is often present in a tumor tissue.

The mathematical modeling of oncolytic viruses started with the foundational work of Wodarz and Wodarz and Komarova (Wodarz, 2001; Wodarz and Komarova, 2009; Komarova and Wodarz, 2010). They developed and analysed general systems of ODEs that are based on the biological properties of tumor growth and viral spread. Inspired by their work, Tian (2011) introduced an immune compartment into the OV model and he demonstrated that virotherapy success strongly depends on the burst size of the oncolytic virus. In (Pooladvand et al., 2021), the OV model was put into a spatial context by adding diffusion terms. They found that virotherapy often falls short as a standalone treatment, due to heterogeneous viral spread patterns in cancer tissue. Their work inspired the model by Baabdulla and Hillen (2024); Bansod and Hillen (2026), who developed a diffusion model of OVT and extensively studied the temporal and spatial interactions of the virus, and the cancer cells, but without considering the effect of the immune system. The introduction of oncolytic viruses, however, triggers pro-inflammatory signaling within the tumor. In Eftimie et al. (2011), an ODE-based model with two immune compartments was developed. Their model successfully replicated experimental tumor growth and immune response patterns, and showed the existence of multistability and multi-instability dynamics. In the work by Storey et al. (2020), they noted that the innate immune system often acts too rapidly, clearing the virus before it can adequately infect tumor cells and activate a strong anti-tumor immune response. Consequently, OVT by itself frequently fails to eliminate tumors. This observation motivated Mohammadnejad and Hillen (2026) to perform treatment optimization analysis on an immune-OVT model, including combinations with immune activators or suppressors and CAR-T cell therapy. Further discussion on the many contributions to this field can be found in the extensive Introduction of Baabdulla and Hillen (2024).

Once the virus has been introduced, it needs to spread over the entire cancer population to infect as many cells as possible. A standard way to analyze such an invasion profile is a travelling wave analysis. (Aronson and Weinberger, 1975, 1978; Weinberger, 1982). We find the existence of *stacked waves*, which are invasion fronts between different equilibria, which invade at different speeds. Feinberg and Terman (1991) showed that if there are multiple distinct, locally stable **equilibrium** points and all of the equilibria are nondegenerate, then there must **exist** a wave train which connects them. Later Roquejoffre et al. (1996) discussed the convergence of the traveling wave fronts of a monotone parabolic system toward stacked families of waves. Since then, the existence of stacked waves has been studied by different authors (Iida et al., 2011; Hamelin et al., 2022; Du and Wu, 2018; Ducrot et al., 2019; Liu et al., 2021), who all showed the existence of waves that can invade with different profiles and different speeds, for various reaction-diffusion type models.

In this work, Section 2 begins with the construction of a cancer–virus–immune model to analyze

the complex dynamics that may arise during oncolytic virotherapy (OVT). Inspired by Baabdulla and Hillen (2024) and Al-Tuwairqi et al. (2020), we extend the OV model of Baabdulla and Hillen (2024) by incorporating an immune cell population. This allows us to investigate the fundamental interactions between the immune response, the tumor, and the oncolytic virus.

In Section 3, we first study the spatially homogeneous model, excluding diffusion terms, and conduct a qualitative analysis of the system and its steady states. We examine the stability of the equilibrium points and analyze how the Hopf bifurcation observed in the OV model of Baabdulla and Hillen (2024) is influenced by the presence of the immune response. Our results indicate that a sufficiently strong immune response suppresses oscillatory dynamics. We then perform a comprehensive sensitivity analysis of all model parameters, which reveals that the immune growth and decay parameters have the greatest impact on treatment outcomes.

In Section 4, we place the OV–immune model into a spatial framework by incorporating diffusion terms. Of particular interest are the invasion speeds of the system. The travelling wave analysis reveals an unexpected phenomenon: rather than converging to a single travelling front, the solution components form stacked fronts that propagate at different speeds. Faster components move ahead, while slower ones lag behind. Specifically, we observe an initial invasion front in which cancer cells invade healthy tissue. This is followed by a viral wave that catches up with the tumor front. Finally, an immune response wave emerges, **which may propagate sufficiently rapidly to catch up with the preceding fronts.** Instead of a single coherent wave, the system exhibits a layered structure of stacked travelling fronts. We close with a Conclusion in Section 5.

2 The mathematical model

We incorporate the population of effector immune cells Y into the oncolytic virus model of (Baabdulla and Hillen, 2024), so our proposed model is given by

$$\begin{aligned}
\frac{dC}{dt} &= D_C\Delta C + rC\left(1-\frac{C+I}{L}\right) - \beta CV - p_1 YC \\
\frac{dI}{dt} &= D_I\Delta I + \beta CV - \alpha I - q_1 YI \\
\frac{dV}{dt} &= D_V\Delta V + \alpha bI - \omega V - s_1 VY \\
\frac{dY}{dt} &= D_Y\Delta Y + \eta_1 YC + \zeta_1 YI - \xi_1 Y
\end{aligned} \tag{1}$$

where $C(x,t), I(x,t), V(x,t)$, and $Y(x,t)$ represent the populations of uninfected cancer cells, infected cancer cells, free virus particles, and immune cells respectively. Their corresponding diffusion coefficients are denoted by D_C, D_I, D_V, and D_Y. We have the Laplace operator Δ as the sum of all second-order derivatives. **In the equation for uninfected cancer cells, the logistic growth term describes cancer cell proliferation in the absence of therapy, with growth rate r and carrying capacity L.** The interactions between tumor cells and virus particles are modeled by the mass action term βCV where cancer cells become infected by the virus at a rate β. The second term in this equation accounts for the increase in the infected cells. The lysis of these infected cells is

Variable	Value	Unit	Description
C_0	10^6	cells/mm^3	Initial density of uninfected cells
I_0	0	cells/mm^3	Initial density of infected ells
V_0	1.9×10^{10}	viruses/mm^3	Initial density of virus particles
Y_0	10^4	units/mm^3	Initial density of immune cells

Table 1: Initial conditions for model 1 variables.

represented by the third term at the rate α. In the third equation, b represents the burst size of the virus in infected cells, while ω denotes the viral clearance rate. The effect of the immune response on cancer cells, infected cancer cells, and the virus is modeled through standard mass action terms with rates p_1, q_1, and s_1, respectively. We do not include a growth term for infected cells, since **oncolytic viruses are designed to lyse host cells following infection** (Coates et al., 2025; Shalhout et al., 2023). The last equation of this system describes the stimulation of the immune system by both uninfected and infected cancer cells with rates η_1 and ζ_1, respectively. The immune cell are cleared at rate ξ_1 as denoted in the last term.

We consider the system (1) with the initial conditions:

$$C(0) = N_0 > 0, \quad I(0) = I_0 = 0, \quad V(0) = V_0 > 0, \quad Y(0) = Y_0 > 0,$$

either on an unbounded domain $\mathbb{R}^n$, or on a bounded domain Ω with homogeneous Neumann boundary conditions on $\partial\Omega$

$$n \cdot \nabla C = n \cdot \nabla I = n \cdot \nabla V = n \cdot \nabla Y = 0. \quad \forall x \in \partial\Omega.$$

Based on the work of Lodish (2013) and Pooladvand et al. (2021), we assume that the carrying capacity of a solid tumor with radius 1 mm is approximately $L = 10^6$ cells per mm^3. We adopted this value as the initial density of uninfected tumor cells, together with the initial adenovirus load $V_0 = 1.9 \times 10^{10}$ virions per mm^3, as reported in the model of Baabdulla and Hillen (2024), where these parameters were associated with a reovirus in the context of breast cancer. We estimated the initial density of the immune cells to be 10^4.

The role of spatial structure in the model is particularly relevant in this model. As already noted in the early oncolytic virus models of Wodarz (2001), different spatial invasion patterns can arise, and these may lead to markedly different clinical outcomes. Our analysis further demonstrates that variations in the invasion speeds of the stacked travelling waves **may substantially influence the overall treatment response.**

2.1 Non-dimensionalization

To simplify our analysis, we apply the non-dimensionalization method by considering

$$\hat{C} = \frac{C}{L}, \quad \hat{I} = \frac{I}{L}, \quad \hat{V} = \frac{\beta V}{r}, \quad \hat{Y} = \frac{Y}{L}, \quad \hat{t} = rt, \text{ and } \quad x = x\sqrt{\frac{r}{D_V}}. \tag{2}$$

	Description	Baseline Value	Units	Source
D_C	Diffusion coefficient of susceptible cells	0.006	mm^2 per day	[BH], [PL]
D_I	Diffusion coefficient of infected cells	0.006	mm^2 per day	[BH], [PL]
D_V	Diffusion coefficient of virus	0.24	mm^2 per day	[BH], [PL]
D_Y	Diffusion coefficient of immune cells	0.006	mm^2 per day	[BH], [PL]
r	Growth rate of uninfected tumor	0.3 $(0.12 - 1.2)$	$(\text{day})^{-1}$	[BH], [ST]
L	Tumor carrying capacity	$10^6(10^6 - 10^8)$	cells	[BH], [PL]
β	Infection rate of uninfected tumor cells by OV	1.5×10^{-9}	$(\text{viruses .day})^{-1}$	[BH], [PL]
p_1	Killing rate of tumor by immune cells	0.48×10^{-6}	$(\text{cell .day})^{-1}$	[AT] (estimated)
α	Death rate of infected tumor cells	1 $(0.5 - 2.6)$	$(\text{day})^{-1}$	[BH], [ST]
q_1	Killing rate of infected tumor by tumor-specific immune	0.63×10^{-6}	$(\text{cell .day})^{-1}$	[AT] (estimated)
b	Viruses released by lysed infected tumor cell	3500 $(1000 - 100,000)$	$(\text{viruses})(\text{cell})^{-1}$	[BH], [PL]
ω	Virus-induced immune response rate	4 $(0.024 - 24)$	$(\text{day})^{-1}$	[BH], [PL]
s_1	Killing rate of virus by immune cells	0.21×10^{-6}	$(\text{cell .day})^{-1}$	[AT] (estimated)
η_1	Stimulation of immune response by uninfected cells	3.84×10^{-7}	$(\text{cell .day})^{-1}$	[AT] (estimated)
ζ_1	Stimulation of immune response by infected cells	7.8×10^{-7}	$(\text{cell .day})^{-1}$	[AT] (estimated)
ξ_1	clearance rate of immune cells	0.036	$(\text{day})^{-1}$	[AT]

Table 2: Baseline parameter values for the base model (1) and their corresponding references [BH] = Baabdulla and Hillen (2024) [AT] = Al-Tuwairqi et al. (2020) [ST]= Storey et al. (2020) [PL]= Pooladvand et al. (2021).

After dropping the hats, we obtain the non-dimensionalized system:

$$\begin{aligned} \frac{dC}{dt} &= D_c\Delta C + C\left(1 - C - I\right) - CV - pYC \\ \frac{dI}{dt} &= D_i\Delta I + CV - \lambda I - qYI \\ \frac{dV}{dt} &= \Delta V + \theta I - aV - sVY \\ \frac{dY}{dt} &= D_y\Delta Y + \eta YC + \zeta YI - \xi Y \end{aligned} \tag{3}$$

where

$$p = \frac{p_1 L}{r}, \quad q = \frac{q_1 L}{r}, \quad s = \frac{s_1 L}{r} \quad a = \frac{\alpha}{r}, \quad \theta = \frac{\alpha b \beta L}{r^2}, \quad \gamma = \frac{\omega}{r}, \quad \eta = \frac{\eta_1 L}{r},$$
$$\zeta = \frac{\zeta_1 L}{r}, \quad \xi = \frac{\xi_1}{r},$$

and

$$D_c = \frac{D_C}{D_V}, \quad D_i = \frac{D_I}{D_V}, \quad D_y = \frac{D_Y}{D_V}.$$

The parameter values used in our simulations are taken from various sources and are listed, before and after nondimensionalization, in Tables 2 and 3, respectively. A detailed explanation of these parameter choices is provided in the Appendix, Section A.

Parameter	Description	Baseline Value
D_c	Diffusion coefficient of susceptible cells	0.025
D_i	Diffusion coefficient of infected cells	0.025
D_y	Diffusion coefficient of immune cells	0.025
θ	Effective viral production rate	58.33
p	Killing rate of tumor by immune cells	1.6
q	Killing rate of infected tumor by tumor-specific immune cells	2.1
a	infected death rate	3.33
γ	Virus clearance rate	13.33
s	Killing rate of virus by immune cells	0.71
η	Stimulation of immune response by uninfected cells	1.28
ζ	Stimulation of immune response by infected cells	2.6
ξ	clearance rate of immune cells	0.16

Table 3: Parameters of the non-dimensional model (3).

3 Analysis of the Kinetic Part

We first examine the aforementioned model in the spatially homogeneous scenario, where all spatial dependencies are disregarded. Consequently, the system governing $C(t), I(t), V(t), Y(t)$ becomes:

$$\begin{aligned} \frac{dC}{dt} &= C\left(1-C-I\right)-CV-pYC \\ \frac{dI}{dt} &= CV-aI-qYI \\ \frac{dV}{dt} &= \theta I-\gamma V-sVY \\ \frac{dY}{dt} &= \eta YC+\zeta YI-\xi Y. \end{aligned} \tag{4}$$

System (4) has five equilibria: The "disease-free" equilibrium is $E_0 = (C, I, V, Y) = (0, 0, 0, 0)$, the "cancer-only" equilibrium is given by $E_1 = (1, 0, 0, 0)$, and the "immune-free" equilibrium has the coordinates

$$E_2 = (C_2, I_2, V_2, 0) = \left(\frac{a\gamma}{\theta}, \frac{\gamma(\theta - a\gamma)}{\theta(\gamma+\theta)}, \frac{\theta - a\gamma}{\gamma+\theta}, 0\right), \tag{5}$$

which is biologically relevant if and only if $\theta > a\gamma$. We denote the "immune-cancer-only" equilibrium as

$$E_3 = (C_3, 0, 0, Y_3) = \left(\frac{\xi}{\eta}, 0, 0, \frac{1-\frac{\xi}{\eta}}{p}\right), \tag{6}$$

which is biologically relevant if and only if $\eta > \xi$. The "coexistence" equilibrium $E_4 = \left(\tilde{C}, \tilde{I}, \tilde{V}, \tilde{Y}\right)$ is a nonzero coexistence point that cannot be expressed explicitly due to complexity. To demonstrate the existence of E_4 we simplify the system (4) by assuming that all populations are nonzero. First, we set

all four equations to zero. From the fourth equation of system (4), we cancel Y, isolate C, and obtain

$$C = -\frac{\zeta}{\eta} I + \frac{\xi}{\eta}. \tag{7}$$

From the second equation of system (4), isolating I gives

$$I = \frac{\gamma}{\theta} V + \frac{sYV}{\theta}. \tag{8}$$

Substituting (8) into (7), we rewrite C as

$$C = \frac{\xi}{\eta} - \frac{\zeta\gamma}{\eta\theta} V - \frac{\zeta s}{\eta\theta} YV. \tag{9}$$

Next, cancelling C from the first equation of system (4) yields

$$1 - C - I - V - pY = 0. \tag{10}$$

By substituting (8) and (9) into (10), we obtain an equation in terms of two variables, V and Y:

$$1 - \frac{\xi}{\eta} + \frac{\zeta\gamma V}{\eta\theta} + \frac{\zeta sYV}{\eta\theta} - \frac{\gamma}{\theta} V - \frac{sYV}{\theta} - V - pY = 0. \tag{11}$$

Similarly, substituting (8) and (9) into the third equation of the system (4) gives another relation in V and Y. Cancelling V from both sides yields

$$\frac{\xi}{\eta} - \frac{\zeta\gamma V}{\eta\theta} - \frac{\zeta sYV}{\eta\theta} - \frac{a\gamma}{\theta} - \frac{asY}{\theta} - \frac{q\gamma Y}{\theta} - \frac{qsY^2}{\theta} = 0. \tag{12}$$

Equations (11) and (12) form a system with two variables. Numerical solutions indicate that this system admits at least one positive solution, establishing the existence of the coexistence point E_4. However, due to the complexity of the equations, explicit expressions cannot be derived, and uniqueness cannot be demonstrated analytically.

The equilibrium points E_0, E_1, and E_2 correspond to the three equilibria of the OV model of Baabdulla and Hillen (2024) and their stability turns out to be the same as in the simpler model. In the E_2 state, the tumor persists while the immune population has been driven to extinction. In the equilibrium point E_3, both infected cells and free viruses tend to zero. **This outcome is caused by a sufficiently strong immune response to the presence of viruses and infected cells; consequently, immune cells mediate viral clearance and eliminate infected cells while tumor cells persist.**

A useful measure in disease modeling is the basic reproduction number R_0 (Roberts, 2007; Heesterbeek, 2002; Heffernan et al., 2005). The basic reproduction number refers to the expected number of secondary infections resulting from a single primary infection. In virotherapy, primary and secondary infections refer to individual tumor cells infected by the oncovirus. Using the framework of van den Driessche and Watmough (2002), we compute the reproduction number as spectral radius of the next generation matrix FG^{-1} (Bianchi et al., 2019). The operators $\mathcal{F}$ and $\mathcal{G}$ describe new infections and

transitions between infected compartments, respectively. The matrices F and G are the corresponding linearizations at the disease free equilibrium E_1. From model (4) have:

$$\mathcal{F} = \begin{bmatrix} 0 \\ CV \\ \theta I \\ 0 \end{bmatrix}, \quad \mathcal{G} = \begin{bmatrix} (1 - C - I)\, C - CV - pYC \\ aI + qYI \\ \gamma V + sVY \\ -\eta YC - \zeta YI + \xi Y \end{bmatrix},$$

Here, the two infected classes are the infected cancer cells I and the free virus particles V, both of which contribute to the generation of new infections. The virus free equilibrium is $(C, I, V, Y) = (1, 0, 0, 0)$ because in the absence of infection the tumor reaches its carrying capacity and immune cells vanish. We linearize $\mathcal{F}$ and $\mathcal{G}$ at this equilibrium E_1. The two infected classes are the infected cancer cells I and the virus V. To find the basic reproduction number we can focus on **the corresponding 2×2 submatrices of the linearized system.** So we have

$$F = \begin{bmatrix} 0 & 1 \\ \theta & 0 \end{bmatrix}, \quad \text{and} \quad G = \begin{bmatrix} a & 0 \\ 0 & \gamma \end{bmatrix}.$$

The basic reproduction number is given by the spectral radius of FG^{-1} (van den Driessche and Watmough, 2002), which is

$$R_0 = \rho(FG^{-1}) = \sqrt{\frac{\theta}{a\gamma}}. \tag{13}$$

This shows that E_1 is only stable when $R_0 < 1$ which is equivalent to when $\theta < a\gamma$.

We summarize the stability in the following result.

Theorem 1. *For our base immune-oncolytic virus model (4), we have the following results:*

- *The basic reproduction number R_0 is given by $\sqrt{\frac{\theta}{a\gamma}}$,*
- *E_0 is always a saddle, and when the basic reproduction number R_0 is less than one or equivalently $\theta < a\gamma$, E_1 is locally asymptotically stable and E_2 is not biologically relevant,*
- *When R_0 is greater than one, E_1 becomes unstable and the coexistence steady state arises through a transcritical bifurcation at $\theta_t = a\gamma$, where E_2 becomes biologically relevant,*
- *E_2 is locally asymptotically stable when*

$$\theta > \frac{\gamma\left(\zeta + \eta a - \xi\right) + \gamma\sqrt{\left(\zeta + \eta a - \xi\right)^2 + 4a\xi(\zeta - \xi)}}{2\xi}, \tag{14}$$

and

$$\theta > \theta_t = a\gamma \quad \textit{and} \quad \kappa(\theta) > 0, \tag{15}$$

where

$$\kappa(\theta) = -\theta^3 + m\theta^2 + \gamma m\theta + a\gamma^3, \quad \textit{and} \quad m = (a+\gamma)^2 + a(\gamma+1).$$

- *There exists a bifurcation value $\theta_H > \theta_t$ with $\kappa(\theta_H) = 0$ such that the system undergoes a Hopf bifurcation at E_2,*
- *E_3 is locally asymptotically stable only if we have*

$$\frac{\eta}{\xi}\left(a + q\left(\frac{1-\frac{\xi}{\eta}}{p}\right)\right)\left(\gamma + s\left(\frac{1-\frac{\xi}{\eta}}{p}\right)\right) > \theta. \tag{16}$$

Proof. The Jacobian of the system (4) at a general equilibrium $(\bar{C}, \bar{I}, \bar{V}, \bar{Y})$ is given by

$$\begin{bmatrix} 1-2\bar{C}-\bar{I}-\bar{V}-p\bar{Y} & -\bar{C} & -\bar{C} & -p\bar{C} \\ \bar{V} & -a-q\bar{Y} & \bar{C} & -q\bar{I} \\ 0 & \theta & -\gamma-s\bar{Y} & -s\bar{V} \\ \eta\bar{Y} & \zeta\bar{Y} & 0 & \eta\bar{C}+\zeta\bar{I}-\xi \end{bmatrix}. \tag{17}$$

Part 1. At $E_0 = (0,0,0,0)$, (17) reduces to

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -a & 0 & 0 \\ 0 & \theta & -\gamma & 0 \\ 0 & 0 & 0 & -\xi \end{bmatrix},$$

which shows that E_0 is a saddle.

Part 2. At $E_1 = (1,0,0,0)$, the Jacobian matrix (17) reduces to

$$\begin{bmatrix} -1 & -1 & -1 & -p \\ 0 & -a & 1 & 0 \\ 0 & \theta & -\gamma & 0 \\ 0 & 0 & 0 & \eta-\xi \end{bmatrix}.$$

The characteristic equation is given by

$$(\lambda+1)(\lambda+\eta+\xi)((a+\lambda)(\gamma+\lambda)-\theta) = 0.$$

Thus, the eigenvalues are given by $\lambda_1 = -1$, $\lambda_2 = \eta - \xi$ and

$$\lambda_{3,4} = \frac{-(a+\gamma) \pm \sqrt{(a-\gamma)^2 + 4\theta}}{2}.$$

This gives that E_1 is only stable if $\xi > \eta$ and $(\gamma+a)^2 > (\gamma-a)^2 + 4\theta$. The second condition is equivalent to $1 > \frac{\theta}{\gamma a}$ or $1 > R_0$.

So far we have shown that E_0 is always a saddle and when the condition $\dfrac{a\gamma}{\theta} < 1$ is satisfied, E_1 is locally asymptotically stable and E_2 is not biologically relevant. However, when $a\gamma > \theta$, E_1 becomes

unstable, and E_2 becomes biologically relevant.

Part 3. To analyze the stability of E_2, we set $C_2 = \dfrac{a\gamma}{\theta}$, so we rewrite E_2 as

$$\left(C_2, \frac{\gamma(1-C_2)}{\theta+\gamma}, \frac{\theta(1-C_2)}{\theta+\gamma}, 0\right).$$

We have that the Jacobian matrix (17) at this point is given by

$$\begin{bmatrix} -C_2 & -C_2 & -C_2 & -pC_2 \\ \dfrac{\theta(1-C_2)}{\theta+\gamma} & -a & C_2 & -\dfrac{q\gamma(1-C_2)}{\theta+\gamma} \\ 0 & \theta & -\gamma & -\dfrac{s\theta(1-C_2)}{\theta+\gamma} \\ 0 & 0 & 0 & \eta C_2 + \dfrac{\zeta\gamma(1-C_2)}{\theta+\gamma} - \xi \end{bmatrix}, \tag{18}$$

and so the characteristic equation is given by

$$\left(\eta C_2 + \frac{\zeta\gamma(1-C_2)}{\theta+\gamma} - \xi - \lambda\right) \begin{vmatrix} -C_2-\lambda & -C_2 & -C_2 \\ \dfrac{\theta(1-C_2)}{\theta+\gamma} & -a-\lambda & C_2 \\ 0 & \theta & -\gamma-\lambda \end{vmatrix} = 0. \tag{19}$$

We rewrite this equations as

$$\left(\lambda + \xi - \eta C_2 - \frac{\zeta\gamma(1-C_2)}{\theta+\gamma}\right)\left(\lambda^3 + P_2\lambda^2 + P_1\lambda + P_0\right) = 0, \tag{20}$$

where

$$P_2 = C_2 + a + \gamma, \quad P_1 = C_2\left(a + \gamma + \frac{\theta\,(1-C_2)}{\theta+\gamma}\right), \text{ and } \quad P_0 = \theta C_2\,(1-C_2)\,. \tag{21}$$

To establish that E_2 is locally asymptotically stable, we need to show that we have negative real eigenvalues or complex eigenvalues with negative real parts. We first disregard the first factor and focus on the eigenvalues associated with the second factor in (20). We then analyze this term to determine the signs of the corresponding eigenvalues. In particular, we aim to show that the polynomial

$$\lambda^3 + P_2\lambda^2 + P_1\lambda + P_0 = 0,$$

has only real negative eigenvalues, or complex eigenvalues with negative real parts, provided that $\theta > a\gamma$ and $\kappa(\theta) > 0$ where $\kappa(\theta) = -\theta^3 + m\theta^2 + \gamma m\theta + a\gamma^3$ and $m = (a+\gamma)^2 + a(\gamma+1)$. **To establish this result, we apply the Routh–Hurwitz criterion (Edelstein-Keshet, 2005 - 1988). Based on this criterion, the third-order polynomial**

$$\lambda^3 + P_2\lambda^2 + P_1\lambda + P_0 = 0$$

has all eigenvalues with negative real parts if the following conditions hold:

$$P_0 > 0, \quad P_2 > 0, \quad \text{and} \quad P_2P_1 - P_0 > 0. \tag{22}$$

Since we assume $C_2 > 1$, it follows that $P_2 > 0$ and $P_0 > 0$. To verify the remaining condition, we substitute the expressions into $P_2P_1 - P_0$. Using the identity $C_2 = \frac{a\gamma}{\theta}$, we obtain

$$P_2P_1 - P_0 = (C_2 + a + \gamma)\left(C_2\left(a + \gamma + \frac{\theta\,(1 - C_2)}{\theta + \gamma}\right)\right) - \theta C_2\,(1 - C_2) \tag{23}$$

$$= C_2\left[(a+\gamma)(C_2 + a + \gamma) + \frac{\theta(1 - C_2)(C_2 + a - \theta)}{\theta + \gamma}\right] \tag{24}$$

$$= \frac{a\gamma}{\theta^2(\theta + \gamma)}k(\theta) \tag{25}$$

where

$$\kappa(\theta) = -\theta^3 + m\theta^2 + \gamma m\theta + a\gamma^3 \quad \text{and,} \quad m = (a+\gamma)^2 + a(\gamma + 1) > 0. \tag{26}$$

Since $\dfrac{a\gamma}{\theta^2(\theta + \gamma)} > 0$, to have the condition $P_2P_1 - P_0 > 0$ satisfied, we should have

$$\kappa(\theta) > 0. \tag{27}$$

Now, we need to analyze the first term in (20) to find all the conditions under which E_2 is locally asymptotically stable. To have this, we need

$$\lambda_4 = \eta C_2 + \frac{\zeta\gamma(1 - C_2)}{\theta + \gamma} - \xi < 0.$$

We rewrite this as

$$\frac{\eta a\gamma}{\theta} + \frac{\zeta\gamma(1 - C_2)}{\theta + \gamma} - \xi = \frac{\eta a\gamma\theta\gamma^2 a\eta + \zeta\gamma\theta - \zeta\gamma^2 a}{\theta(\theta + \gamma)} - \xi < 0.$$

This gives us

$$\xi\theta^2 + \theta\,(\xi\gamma - \zeta\gamma - \eta a\gamma) + a\gamma^2(\zeta - \xi) < 0,$$

and so to have the condition (27) satisfied, we need

$$\theta > \frac{\gamma\,(\zeta + \eta a - \xi) + \gamma\sqrt{(\zeta + \eta a - \xi)^2 + 4a\xi(\zeta - \xi)}}{2\xi}. \tag{28}$$

We showed that the condition (22) is satisfied, thus to have E_2 locally asymptotically stable, we need the inequality (28) to be satisfied as well.

Part 4. We next show that the system (4) undergoes a Hopf bifurcation at θ_H for

some $\theta_H > \theta_t$ around the equilibrium point E_2. We therefore consider the characteristic equation (20). Since the first factor is always real-valued, it cannot generate a Hopf bifurcation; consequently, we restrict our attention to the second factor.

$$\lambda^3 + P_2\lambda^2 + P_1\lambda + P_0 = 0$$

with P_2, P_1, and P_0 values given in (21). We define $H(\theta)$ as

$$H(\theta) \coloneqq P_2P_1 - P_0 = \frac{a\gamma}{\theta^2(a+\gamma)}\kappa(\theta) \tag{29}$$

with $\kappa(\theta)$ defined in (26). We apply Liu's criterion (Liu, 1994; Jahedi et al., 2024), which states that, assuming $P_0 \neq 0$, a Hopf bifurcation occurs at $\theta = \theta_H$ for system (3) whenever

$$H(\theta_H) = 0 \quad \text{and} \quad \left.\frac{dH}{d\theta}\right|_{\theta=\theta_H} \neq 0.$$

Since the multiplicative factor of $H(\theta)$, namely $a\gamma$, is nonzero, it follows that $H(\theta) = 0$ if and only if $\kappa(\theta) = 0$. By Descartes' rule of signs (Curtiss, 1918), the third order polynomial $\kappa(\theta)$ admits exactly one positive real root, since there is only one sign variation among its coefficients. Considering $\kappa(-\theta)$, there are two sign changes, implying that $\kappa(\theta)$ admits either zero or two negative real roots. Therfore, there exists a unique positive solution $\theta = \theta_H$ such that

$$H(\theta_H) = \kappa(\theta_H) = 0.$$

Moreover, since θ_H is the unique positive root of $\kappa(\theta)$ and the leading-order term of $\kappa(\theta)$ is negative, we conclude that $\kappa(\theta) < 0$ for all $\theta > \theta_H$. Consequently, since $H(\theta)$ in Equation (29) has a positive multiple, $a\gamma > 0$, we also have $H(\theta) < 0$ for all $\theta > \theta_H$. Finally, from (27), we know that $\kappa(\theta) > 0$, which implies $\theta_t < \theta_H$.

Part 5. To analyze the stability of equilibria $E_3 = (C_3, 0, 0, Y_3)$, we consider the Jacobian matrix given by

$$\begin{bmatrix} -C_3 & -C_3 & -C_3 & -pC_3 \\ 0 & -a - qY_3 & C_3 & 0 \\ 0 & \theta & -\gamma - sY_3 & 0 \\ \eta Y_3 & \zeta Y_3 & 0 & 0 \end{bmatrix}. \tag{30}$$

The characteristic equation is given by

$$\begin{aligned} &\begin{bmatrix} -C_3 - \lambda & -C_3 & -C_3 & -pC_3 \\ 0 & -a - qY_3 - \lambda & C_3 & 0 \\ 0 & \theta & -\gamma - sY_3 - \lambda & 0 \\ \eta Y_3 & \zeta Y_3 & 0 & -\lambda \end{bmatrix} \\ &= \left[(-a - qY_3 - \lambda)(-\gamma - sY_3 - \lambda) - C_3\theta\right]\left[-\lambda(-C_3 - \lambda) + pC_3\eta Y_3\right] = 0. \end{aligned} \tag{31}$$

Considering the cancer-immune equilibria $E_3 = (C_3, 0, 0, Y_3)$ (6), we substitute the values of C_3 and Y_3 into the Equation (31). The second factor gives us

$$\lambda^2 + \frac{\xi}{\eta}\lambda + \xi\left(1 - \frac{\xi}{\eta}\right) = 0. \tag{32}$$

The sum of the roots of this quadratic equation is given by the negative coefficient of the linear term, namely $-\frac{\xi}{\eta}$, which is strictly negative. The product of the roots is given by the constant term, $\xi\left(1 - \frac{\xi}{\eta}\right)$, which is positive since $\xi < \eta$. Therefore, if the quadratic equation admits real roots, both roots are negative. Consequently, the eigenvalues associated with this factor of the characteristic equation (31) are negative.

We proceed similarly for the first factor of (31). Substituting the expressions for C_3 and Y_3 into the first factor of (31) yields

$$\lambda^2 + \lambda\left(a + q\left(\frac{1-\frac{\xi}{\eta}}{p}\right) + \gamma + s\left(\frac{1-\frac{\xi}{\eta}}{p}\right)\right) + \left(a + q\left(\frac{1-\frac{\xi}{\eta}}{p}\right)\right)\left(\gamma + s\left(\frac{1-\frac{\xi}{\eta}}{p}\right)\right) - \theta\frac{\xi}{\eta} = 0. \tag{33}$$

The sum of the roots of this equation is equal to the negative of the coefficient of the linear term, i.e.

$$-\left(a + q\left(\frac{1-\frac{\xi}{\eta}}{p}\right) + \gamma + s\left(\frac{1-\frac{\xi}{\eta}}{p}\right)\right), \tag{34}$$

which has a negative value. If real roots exist, a necessary condition for both to be negative is that their product be positive. Therefore, we require the constant term of equation (33) to be positive, namely,

$$\frac{\eta}{\xi}\left(a + q\left(\frac{1-\frac{\xi}{\eta}}{p}\right)\right)\left(\gamma + s\left(\frac{1-\frac{\xi}{\eta}}{p}\right)\right) > \theta. \tag{35}$$

Hence E_3 is locally asymptotically stable only if we have (16). □

We consider η as a bifurcation parameter and for low virus concentration, we find a transcritical bifurcation in η.

Lemma 2. *Assume the virus concentration V and the infected population I are close to 0. Then by increasing η, we obtain a transcritical bifurcation between E_1 and E_3 at $\eta = \xi$.*

Proof. To study the transcritical bifurcation between the equilibria $E_1 = (1, 0, 0, 0)$, and $E_3 = (C_3, 0, 0, Y_3)$, we set $I = 0$ and $V = 0$. In this case, system (4) reduces to

$$\begin{aligned} C' &= C(1 - C) - pCY, \\ Y' &= \eta CY - \xi Y. \end{aligned} \tag{36}$$

The reduced system admits three steady states: the trivial equilibrium $e_0 = (0, 0)$, the cancer-only equilibrium $e_1 = (1, 0)$, and the coexistence equilibrium $e_2 = (c_3, y_3) = \left(\frac{\xi}{\eta}, \frac{1-\frac{\xi}{\eta}}{p}\right)$. These equilibria

are consistent with those found in the full system (3) when the infected population I and the virus population V are set to zero.
The Jacobian of system (36) at a steady state $(\bar{C}, \bar{Y})$ is

$$J(C,Y) = \begin{pmatrix} 1 - 2\bar{C} - p\bar{Y} & -p\bar{C} \\ \eta\bar{Y} & \eta\bar{C} - \xi \end{pmatrix}. \tag{37}$$

At the trivial equilibrium $(0,0)$, the system is always a saddle. At the cancer-only equilibrium $(1,0)$, the Jacobian is

$$\begin{pmatrix} -1 & -p \\ 0 & \eta - \xi \end{pmatrix}, \tag{38}$$

which shows that $(1,0)$ is stable when $\xi > \eta$. The condition $\xi = \eta$, therefore, defines a bifurcation point.
At the coexistence equilibrium $e_2 = (c_3, y_3)$, the Jacobian becomes

$$\begin{pmatrix} -c_3 & -pc_3 \\ \xi y_3 & 0 \end{pmatrix}. \tag{39}$$

The eigenvalues are given by

$$\lambda_{1,2} = \frac{-c_3 \pm \sqrt{c_3^2 - 4p\xi c_3 y_3}}{2} = \frac{-\frac{\xi}{\eta} \pm \sqrt{\frac{\xi^2}{\eta^2} - 4\frac{\xi^2}{\eta}\left(1 - \frac{\xi}{\eta}\right)}}{2}.$$

For $\xi < \eta$, the real parts of the eigenvalues are negative, while for $\xi > \eta$, the equilibrium e_2 becomes a saddle. Hence, a transcritical bifurcation occurs between e_1 and e_2 at $\xi = \eta$.

□

In Figure 1, we plot the curves

$$\kappa(\theta) = -\theta^3 + m\theta^2 + \gamma m\theta + a\gamma^3, \quad \text{and} \quad m = (a+\gamma)^2 + a(\gamma+1), \tag{40}$$

using the parameters values from Table 3. We have indicated the points $\theta_t = a\gamma$ and the Hopf bifurcation value θ_H on this curve. Their corresponding values are 44.4 and 338.4, respectively. In Figure 2, we illustrate the stability behavior of system 4 for $\theta \in [0, 500]$ and $\eta \in [0, 2]$. The shaded regions, labeled A–F, classify the qualitative behavior of the system 4. Region A (green) corresponds to the stability of E_3, region E (light blue) to the stability of E_2, and region F (red) to the stability of the cancer-only equilibrium E_1. For θ exceeding the Hopf bifurcation threshold θ_H (grey line), stable limit cycles around E_2 emerge, forming region D (dark blue). Region B (yellow) represents the stability of E_4, while region C (orange) indicates numerically observed stable limit cycles around E_4.

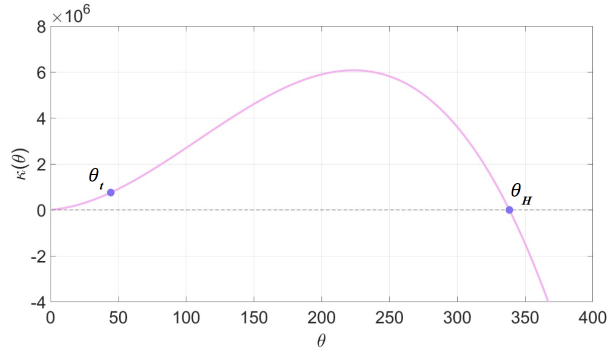


Figure 1: The $\kappa(\theta)$ curve of Equation (26), with the parameter values from Table 3 the bifurcation values $\theta_t = 44.4$ and $\theta_H = 338.45$ are indicated on the curve.

Using inequality (35), we define the curve

$$\theta_1(\eta) = \frac{\eta}{\xi}\left(a + q\left(\frac{1-\frac{\xi}{\eta}}{p}\right)\right)\left(\gamma + s\left(\frac{1-\frac{\xi}{\eta}}{p}\right)\right). \tag{41}$$

For η values above the curve $\theta_1(\eta)$ (green line), equilibrium E_3 is stable (region A). In this regime, the virus is cleared and the system approaches the cancer–immune equilibrium. Consequently, successful virotherapy requires parameter values below $\theta_1(\eta)$. Similarly, inequality (28) yields the curve

$$\theta_2(\eta) = \frac{\gamma\left(\zeta + \eta a - \xi\right) + \gamma\sqrt{\left(\zeta + \eta a - \xi\right)^2 + 4a\xi(\zeta - \xi)}}{2\xi}. \tag{42}$$

For $\theta_2(\eta)$ (blue line), stability changes occur below this curve. Specifically, for $44.4 < \theta < \theta_H = 338.45$, E_2 is stable (region E). When $\theta > \theta_H$, a Hopf bifurcation gives rise to stable limit cycles around E_2 (region D). Between the curves $\theta_1(\eta)$ and $\theta_2(\eta)$, equilibrium E_4 is stable (region B). For sufficiently large θ and η within this region, oscillatory dynamics around E_4 are observed numerically (region C). Since the transition to oscillations cannot be derived analytically, we indicate it with an orange dashed line. The dashed purple line provides a numerical estimate of the gradual transition from oscillations around E_2 to oscillations around E_4. Representative parameter values from each region (A–F) are marked in Figure 2 by blue markers, whose locations indicate the specific value of the parameters (η, θ) selected for the simulations.

Their corresponding trajectories are shown in Figures 3 and 4. These figures display the three-dimensional dynamics of system 4 in the (C, V, Y) phase space; the variable I is omitted since it closely tracks V. For region A, with $(\eta, \theta) = (1.28, 400)$, the system converges to the cancer–immune equilibrium (Figures 3(a,b)). In region B, for $(\eta, \theta) = (1, 400)$, simulations show convergence to a non-zero coexistence state consistent with E_4 (Figures 3(c,d)), even though E_4 is not available in closed form. In region C, $(\eta, \theta) = (0.7, 400)$ produces a stable limit cycle around E_4 (Figures 3(e,f)). Region E is illustrated by $(\eta, \theta) = (0.15, 300)$, where the trajectories converge to the immune-free equilibrium E_2 (Figures 4(a,b)). Increasing θ beyond θ_H while keeping $\eta = 0.15$ (e.g. $\theta = 400$) leads to a stable limit cycle around E_2 (region D; Figures 4(c,d)). Finally, region F corresponds to $(\eta, \theta) = (0, 44.4)$, where

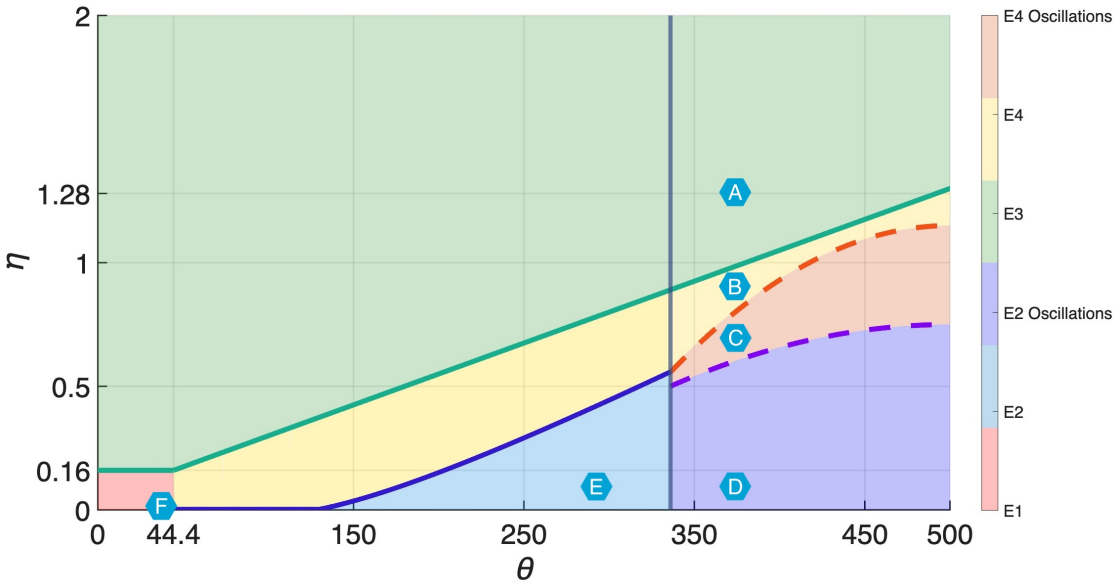


Figure 2: Stability behavior of system (4) over a range of θ and η values. The grey line indicates the Hopf bifurcation value θ_H for the equilibrium E_2. The equations given in (41) and (42) are represented by the green and blue curves respectively. The blue markers highlight parameter values corresponding to the dynamics shown in Figures 3 and 4.

the system converges to the cancer-only equilibrium $E_1 = (1, 0, 0, 0)$ (Figures 4(e,f)). In all parameter ranges examined, we found no evidence of bi-stability. While this does not rigorously exclude its possibility *per se*, it is not expected for this system.

3.1 Sensitivity Analysis

We perform a parameter sensitivity analysis with the base value parameters from Table 3. We use this analysis to identify the parameters that most significantly contribute to treatment efficacy. The relative sensitivity of variable C to the parameter p is defined as (Ingalls, 2013)

$$S(C|p) = \frac{\Delta C/C}{\Delta p/p} = \frac{C(p + \Delta p) - C(p))}{C(p)} \frac{p}{\Delta p}. \tag{43}$$

The relative sensitivity relates the size of a relative perturbation in p to a relative change in C. We vary each parameter by 5% of its value, and we obtain the sensitivity graph as in Figure 5. The sensitivity analysis indicates the importance of the immune system in determining the success of oncolytic virotherapy (OVT). While the parameter θ has a relatively strong influence compared to most other parameters, its effect is less pronounced than that of η and ξ. The model predictions are particularly sensitive to these two parameters. The sensitivity analysis identifies immune stimulation (η), immune exhaustion (ξ), and the effective viral production rate (θ) as the most influential parameters in the model. This suggests that therapeutic strategies designed to modulate these mechanisms

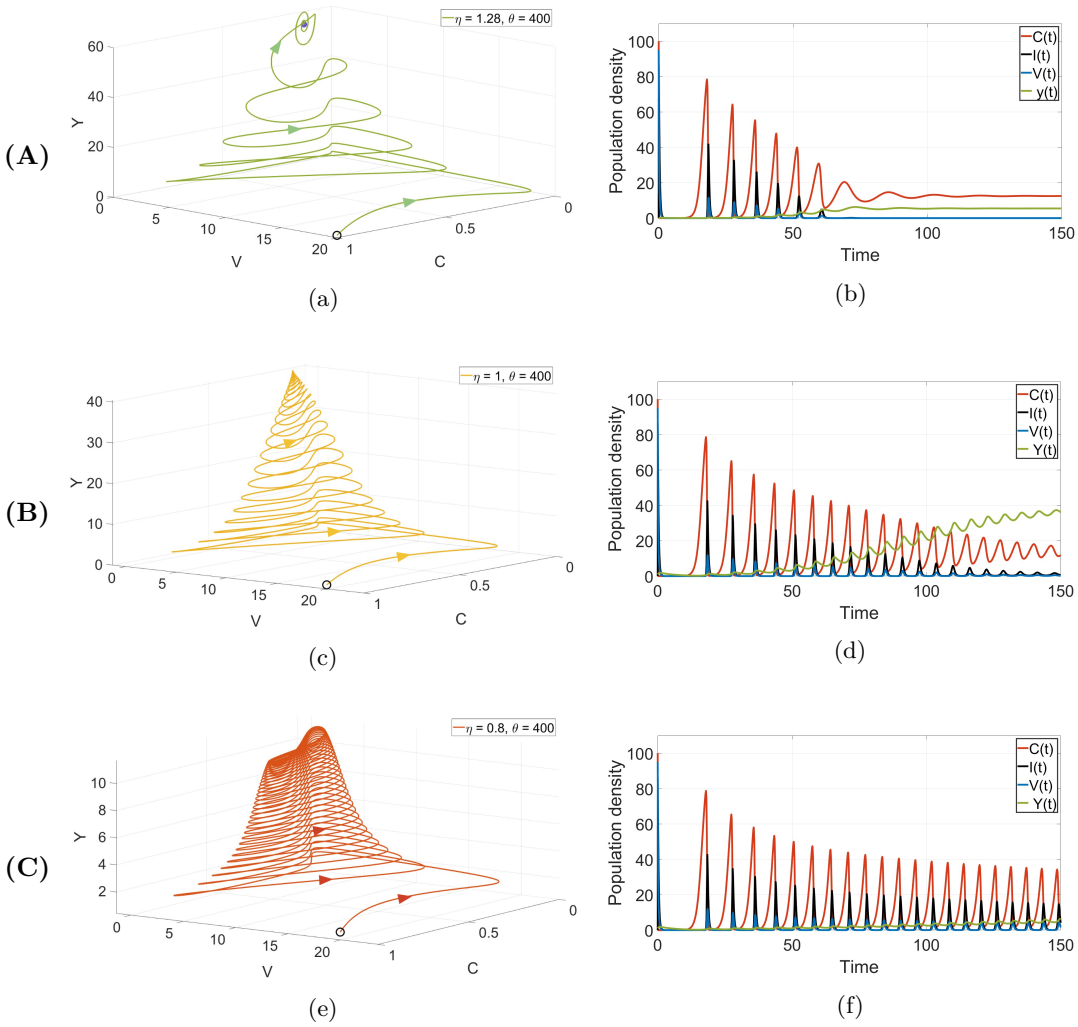


Figure 3: Dynamics of the system (3) for various values of θ and η. The values used for each simulation are shown in the legends of the left-hand plots. The corresponding two-dimensional time dynamics, with parameter values, are indicated on the right. (a) and (b): the equilibrium E_3 is stable. (c) and (d): the equilibrium E_3 is stable, prior to the Hopf bifurcation. (d) and (e): A stable limit cycle emerges around E_4.

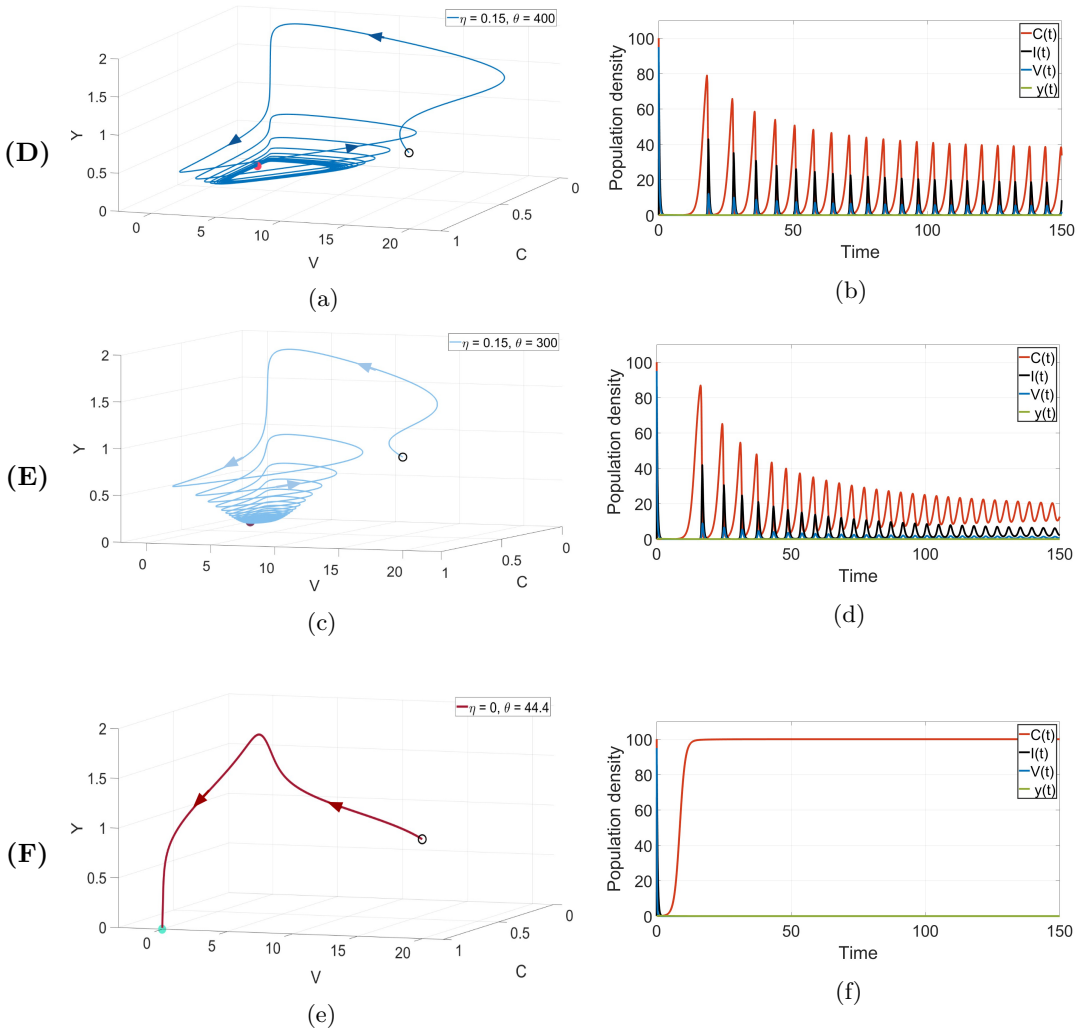


Figure 4: Dynamics of the system (3) for various values of θ and η. The values used for each simulation are shown in the legends of the left-hand plots. The corresponding two-dimensional time dynamics, with parameter values, are indicated on the right. (a) and (b): the equilibrium E_1 is stable. (c) and (d): the equilibrium E_2 is stable for $\theta < \theta_H$, prior to the Hopf bifurcation. (d) and (e): A stable limit cycle emerges around E_2 for $\theta > \theta_H$.

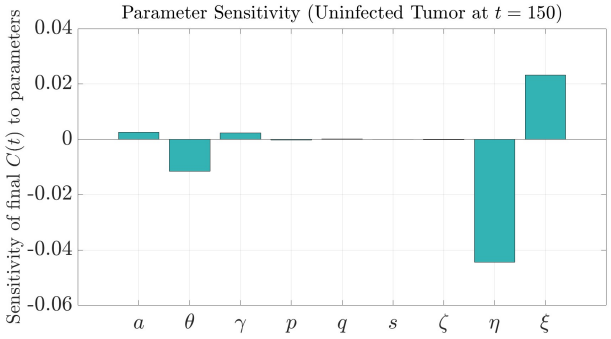


Figure 5: Sensitivity analysis of the final value of $C(t)$ at $t = 150$ under a 5% change in the model parameter values.

may provide the greatest potential for improving treatment outcomes.

These parameters may be modulated through therapeutic approaches such as combining OVT with immunotherapies. The importance of such combination strategies is further supported by the treatment optimization study in Mohammadnejad and Hillen (2026), which identified interventions targeting these mechanisms as effective means of improving treatment outcomes.

4 Travelling Fronts and Stacked Waves

In this section, we come back to the spatial model (3) and analyse the invasion speeds of the various components. The model in one dimension reads

$$\begin{aligned} C_t &= D_c C_{xx} + C\,(1 - C - I) - CV - pYC \\ I_t &= D_i I_{xx} + CV - aI - qYI \\ V_t &= V_{xx} + \theta I - \gamma V - sVY \\ Y_t &= D_y Y_{xx} + \eta YC + \zeta YI - \xi Y, \end{aligned} \tag{44}$$

which we consider on the real line $-\infty < x < \infty$. We first simulate the travelling waves in MATLAB to observe their behavior over time. We choose parameter values from Table 3 and diffusion coefficients as $D_c = D_i = D_y = 0.025$. The initial conditions for I and V are set at E_2, with Y chosen to be equal to a small value, 0.18, for x between 0 and 4 on the left boundary, and zero elsewhere. The uninfected cancer population C is initialized at its carrying capacity up to $x = 30$, and 0 elsewhere, representing the invasion of cancer into the spatial domain. This setup allows us to illustrate the invasion of cancer into the healthy domain $x > 30$. As shown in Figure 6, the leading edge of the wave represents the invasion of uninfected cancer cells (C) into healthy tissue. This is followed—at a higher speed—by the virus and infected cells overtaking the uninfected population, reducing the cancer population, until it reaches the wave front, driving the system toward the steady state E_2, where the immune population is at zero. Further back, at a much slower speed, the immune system begins invading the other three populations. **Over time, the virus and infected cell populations decline to zero, and the**

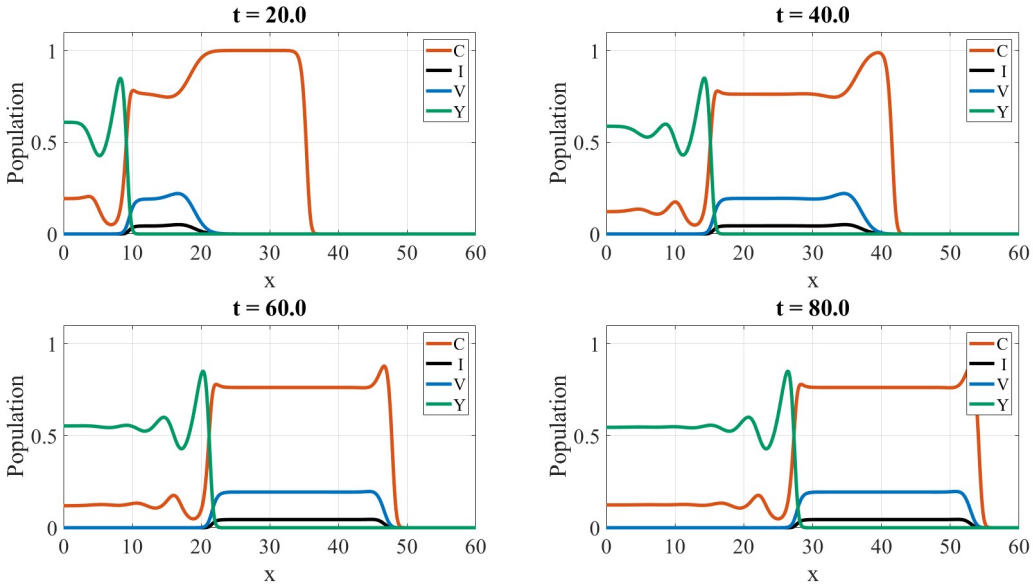


Figure 6: The travelling waves of the four populations invading to the right. The waves are captured at $t = 20,\ 40,\ 60$, and 80.

system approaches a coexistence state with a nonzero cancer population, corresponding to the equilibrium E_3. Additional numerical simulations (not shown) exhibit the same qualitative behavior, suggesting that this dynamical regime persists across a broad range of initial conditions and parameter values. The simulations reveal a sequence of invasion fronts propagating at distinct speeds, corresponding to stacked waves. To analyze this phenomenon, we determine the linear invasion speeds associated with each equilibrium in turn.

Invasion of the Disease-Free Equilibrium E_0: To compute the invasion wave front of the cancer into the disease-free equilibrium, we simply consider the cancer-only model

$$C_t = D_c C_{xx} + C(1 - C),$$

with boundary condition $C(-\infty, t) = 1$ and $C(\infty, t) = 0$. This is the famous Fisher-KPP equation (de Vries et al., 2006) and the invasion speed is well known and given by

$$c_c^* = 2\sqrt{D_c}. \tag{45}$$

Using the parameter values from Table 3, Equation (46) gives us the minimum speed value

$$c_c^* = 2\sqrt{0.025} \approx 0.3162. \tag{46}$$

We show this invasion front in Figure 8a.

Invading the Cancer-Only Equilibrium E_1: To consider the invasion of the virus population

into an established tumor, we linearize the system in one dimension at the homogeneous steady state $(1, 0, 0, 0)$

$$\begin{aligned} C_t &= D_c C_{xx} - C - I - V - pY \\ I_t &= D_i I_{xx} - aI + V \\ V_t &= V_{xx} + \theta I - \gamma V \\ Y_t &= D_y Y_{xx} + \eta Y - \xi Y. \end{aligned} \tag{47}$$

We now make an explicit ansatz of an exponentially decaying self-similar wave solution by setting $z = x - ct$ as

$$(C(x,t), I(x,t), V(x,t), Y(x,t)) = \left(\epsilon_c e^{-\lambda z}, \epsilon_i e^{-\lambda z}, \epsilon_v e^{-\lambda z}, \epsilon_y e^{-\lambda z}\right).$$

Here, c denotes the wave speed, and λ the exponential decay rate at the wave front. Substituting this ansatz into system (47), we get

$$\begin{aligned} c\lambda\epsilon_c e^{-\lambda z} =& D_c\lambda^2\epsilon_c e^{-\lambda z} - \epsilon_c e^{-\lambda z} - \epsilon_i e^{-\lambda z} - \epsilon_v e^{-\lambda z} - p\epsilon_y e^{-\lambda z}, \\ c\lambda\epsilon_i e^{-\lambda z} =& D_i\lambda^2\epsilon_i e^{-\lambda z} - a\epsilon_i e^{-\lambda z} + \epsilon_v e^{-\lambda z}, \\ c\lambda\epsilon_v e^{-\lambda z} =& \lambda^2\epsilon_v e^{-\lambda z} + \theta\epsilon_i e^{-\lambda z} - \gamma\epsilon_v e^{-\lambda z}, \\ c\lambda\epsilon_y e^{-\lambda z} =& D_y\lambda^2\epsilon_c e^{-\lambda z} + (\eta - \xi)\,\epsilon_y e^{-\lambda z}, \end{aligned}$$

which, after simplification, is written as a linear system

$$\begin{bmatrix} D_c\lambda^2 - c\lambda - 1 & -1 & -1 & -p \\ 0 & D_i\lambda^2 - c\lambda - a & 1 & 0 \\ 0 & \theta & \lambda^2 - c\lambda - \gamma & 0 \\ 0 & 0 & 0 & D_y\lambda^2 - c\lambda + \eta - \xi \end{bmatrix} \begin{bmatrix} \epsilon_c \\ \epsilon_i \\ \epsilon_v \\ \epsilon_y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}. \tag{48}$$

The characteristic equation of the matrix (48) is

$$\left(D_c\lambda^2 - c\lambda - 1\right)\left[\left(D_i\lambda^2 - c\lambda - a\right)\left(\lambda^2 - c\lambda - \gamma\right) - \theta\right]\left[\left(D_y\lambda^2 - c\lambda + \eta - \xi\right)\right] = 0. \tag{49}$$

For the first factor of Equation (49), we set

$$\Pi(c, \lambda) = D_c\lambda^2 - c\lambda - 1 = 0. \tag{50}$$

By calculation $\dfrac{d\Pi}{d\lambda} = 0$, isolating for c', and setting it equal to zero, we obtain the minimum value for c. Plugging this value back to the Equation (50) yields $-D_c\lambda^2 - 1 = 0$. For no real value of λ, the equation can be satisfied and it shows that this factor does not give us any suitable minimum wave speed. We now focus on the second factor of Equation (49). We let

$$\Phi(c, \lambda) = \left(D_i\lambda^2 - c\lambda - a\right)\left(\lambda^2 - c\lambda - \gamma\right) - \theta.$$

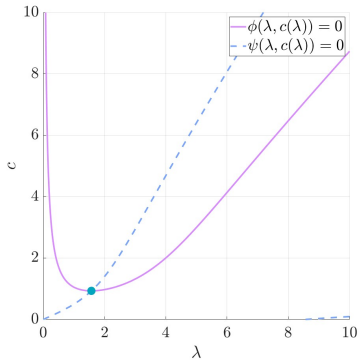


Figure 7: The blue ball corresponds to the minimum decay rate and minimum wave speed of the virus particles.

We set $\frac{d}{d\lambda}\Phi(\lambda, c(\lambda)) = 0$, which is given by

$$(2\lambda D_i - c - c'\lambda)(\lambda^2 - c\lambda - \gamma) + (2\lambda - c - c'\lambda)(D_i\lambda^2 - c\lambda - a) = 0. \tag{51}$$

We solve for c' and we obtain

$$c' = \frac{\Psi(\lambda, c(\lambda))}{\lambda^3(-D_i - 2) + \lambda(a + \gamma)}, \tag{52}$$

where

$$\Psi(\lambda, c(\lambda)) = 4D_i\lambda^3 + 3\lambda^2(-cD_i - c) + 2\lambda(-D_i\gamma + c^2 - a) + (c\gamma + ca).$$

The minimum wave speed c_v^* is the value where $\Phi = 0$ and $\Psi = 0$. We show the functions $\Phi(\theta)$ and $\Psi(\theta)$ in Figure 7. Using the parameter values from Table 3, we numerically solve for the minimum wave speed as

$$c_v^* \approx 0.9297 \quad \text{and} \quad \lambda_v^* \approx 1.5726. \tag{53}$$

This speed corresponds to the invasion of virus and infected cancer cell populations into an established cancer and is illustrated in Figure 8b. It remains to analyze the third term in equation (49). We use the same approach by defining $\phi(\lambda, c)$ as

$$\phi(\lambda; c) = D_y\lambda^2 - c\lambda + \eta - \xi.$$

By setting $\phi(\lambda; c) = 0$, we obtain

$$\lambda = \frac{c \pm \sqrt{c^2 - 4D_y(\eta - \xi)}}{2D_y}. \tag{54}$$

We find the minimum wave speed values $c(\lambda)$ as the minimum value of c for which λ is real. Thus, we

have

$$c_y^* = 2\sqrt{D_y(\eta - \xi)}.$$

Using the parameters from Table (3), this value corresponds to:

$$c_y^* \approx 2\sqrt{0.025(1.28 - 0.16)} \approx 0.3346, \text{ and } \lambda_y^* = 6.692. \tag{55}$$

This speed describes the invasion of immune cells into an established tumor. A simulation is shown in Figure 8c.

Invading the Immune-Free Equilibrium E_2: This case corresponds to the speed with which the immune system invades the immune-free equilibrium state E_2. We linearize (18) at $E_2 = (C_2, I_2, V_2, 0)$.

$$\begin{aligned} C_t &= D_c C_{xx} - C_2 C - C_2 I - C_2 V - pC_2 Y \\ I_t &= D_i I_{xx} + V_2 C - aI + C_2 V - qI_2 Y \\ V_t &= V_{xx} + \theta I - \gamma V - sV_2 Y \\ Y_t &= D_y Y_{xx} + (\eta C_2 + \zeta I_2 - \xi) Y. \end{aligned} \tag{56}$$

Similarly to the previous case, we make an explicit ansatz for an exponentially decaying self-similar wave solution by setting $z = x - ct$ as

$$(C(x,t), I(x,t), V(x,t), Y(x,t)) = \left(\epsilon_c e^{-\lambda z}, \epsilon_i e^{-\lambda z}, \epsilon_v e^{-\lambda z}, \epsilon_y e^{-\lambda z}\right).$$

Substituting this ansatz into system (56) and writing it as a linear system we obtain

$$\begin{bmatrix} D_c\lambda^2 - c\lambda - C_2 & -C_2 & -C_2 & -pC_2 \\ V_2 & D_i\lambda^2 - c\lambda - a & C_2 & -qI_2 \\ 0 & \theta & \lambda^2 - c\lambda - \gamma & -sV_2 \\ 0 & 0 & 0 & D_y\lambda^2 - c\lambda + \eta C_2 + \zeta I_2 - \xi \end{bmatrix} \begin{bmatrix} \epsilon_c \\ \epsilon_i \\ \epsilon_v \\ \epsilon_y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}. \tag{57}$$

The characteristic equation is given by

$$\left(D_y\lambda^2 - c\lambda + \eta C_2 + \zeta I_2 - \xi\right) \begin{vmatrix} D_c\lambda^2 - c\lambda - C_2 & -C_2 & -C_2 \\ V_2 & D_i\lambda^2 - c\lambda - a & C_2 \\ 0 & \theta & \lambda^2 - c\lambda - \gamma \end{vmatrix} = 0 \tag{58}$$

Let us define $\chi(\lambda, c)$ as the first factor

$$\chi(\lambda; c) = D_y\lambda^2 - c\lambda + \eta C_2 + \zeta I_2 - \xi. \tag{59}$$

By setting $\chi(\lambda; c) = 0$, we obtain

$$\lambda = \frac{c \pm \sqrt{c^2 - 4D_y(\eta C_2 + \zeta I_2 - \xi)}}{2D_y} \tag{60}$$

We find the minimum wave speed values $c(\lambda)$, as the minimum value χ for which it has a real solution.

Thus, we have

$$c_a^* = 2\sqrt{D_y(\eta C_2 + \zeta I_2 - \xi)} \tag{61}$$

Using the parameters from Table 3, we have this value corresponding to:

$$c_a^* \approx 0.304 \tag{62}$$

4.1 Numerical Simulations

To compute the speeds of the four travelling waves numerically and compare them with the theoretical values derived above, we initialize each wave in a separate simulation. In Figure 8a, the uninfected cancer population is set at its carrying capacity on the left side of the domain, while all other populations are set to zero. **This configuration generates a cancer invasion front propagating to the right into healthy tissue. The arrow indicates the position of the wave front. Measuring its displacement between $t = 20$ and $t = 40$ gives $\Delta X = 6.20$, corresponding to a numerical wave speed of $c_1^* \approx 0.31215$. This value is consistent with the theoretical prediction $c^* \approx 0.3162$ given in (46).** For the second wave, corresponding to the virus invasion speed into the uninfected cancer population (Fig. 8b), we set the right boundary to the steady state $E_1 = (1, 0, 0, 0)$, while the left boundary is set to the immune-free equilibrium E_2. The arrows indicate the progression of the viral front into the C population, reducing its value and driving the system toward the steady state E_2, where the immune population is absent. From $t = 20$ to $t = 40$, the wave advances by $\Delta X = 18.009$, giving $c_2^* \approx 0.900$, which is in good agreement with the theoretical estimate $c^* \approx 0.930$ (53). For the third wave (Fig. 8c), we simulate the immune system invading the cancer-only equilibrium E_1. Similarly, the arrows show the immune front, which reduces the C population to a nonzero value, consistent with the virus-free steady state E_3. Between $t = 40$ and $t = 80$, the marker advances $\Delta X = 13.415$, giving $c_3^* \approx 0.335$, which is consistent with the theoretical value in $c^* \approx 0.334$ (55). In Figure 8d, we simulate immune-cell invasion into the immune-free equilibrium E_2. Similarly, the arrows show the immune front, which eliminates the V and I populations and reduces the C population to a nonzero value, consistent with the steady state E_3. Between $t = 40$ and $t = 80$, the marker advances by $\Delta X = 12.206$, giving $c_4^* \approx 0.3051$, which matches well with the theoretical value in $c^* \approx 0.304$ (55).

A simulation of the full stacked wave for this example is shown in the earlier Figure 6. Here the virus infection is the fastest wave and it catches up with the growing tumor after some time. The immune response is rather slow, even slower than the cancer invasion front (see Table 4), hence, in this example, the immune response would have a minimal effect on the virus infection of the cancer. **This behavior differs from that reported in earlier studies**, where it was suggested that the immune response removes the virus before it infects all cancer cells.

The wave ordering, however, depends on the model parameters. In particular, increasing the immune-cell diffusion coefficient increases the predicted immune invasion speed. In the baseline simulations, the diffusion coefficients of immune and cancer cells are assumed to be equal (Table 3). If the immune-cell diffusion coefficient is increased

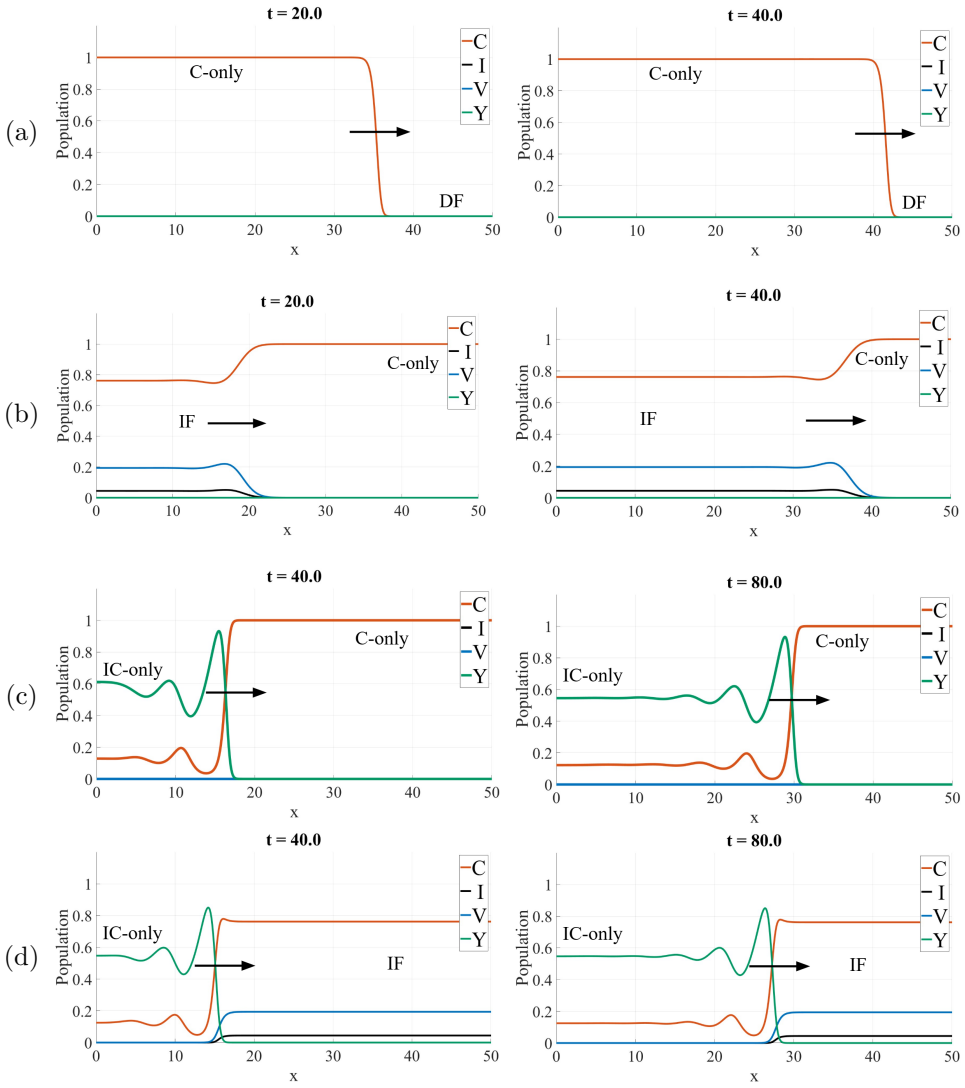


Figure 8: Traveling waves of the different populations at various times: (a) The cancer-only (C-only) wave invading to the right to the disease-free (DF) equilibria. The wave moved from $X_1 = 35.298$ in $t = 20$ to $X_2 = 41.541$ at $t = 20$. (b) The imune-free (IF) equilibria invading the Cancer only (C-only). The wave moved from $X_1 = 18.969$ in $t = 20$ to $X_2 = 36.978$ at $t = 20$. (c) The immune-cancer only (IC-only) invading cancer-only(c-only) equilibria. The wave moved from $X_1 = 16.28$ in $t = 40$ to $X_2 = 29.695$ at $t = 80$. (d) The immune-cancer only (IC-only) invading immune-free(IF) equilibria. The wave moved from $X_1 = 15.048$ in $t = 40$ to $X_2 = 27.254$ at $t = 80$.

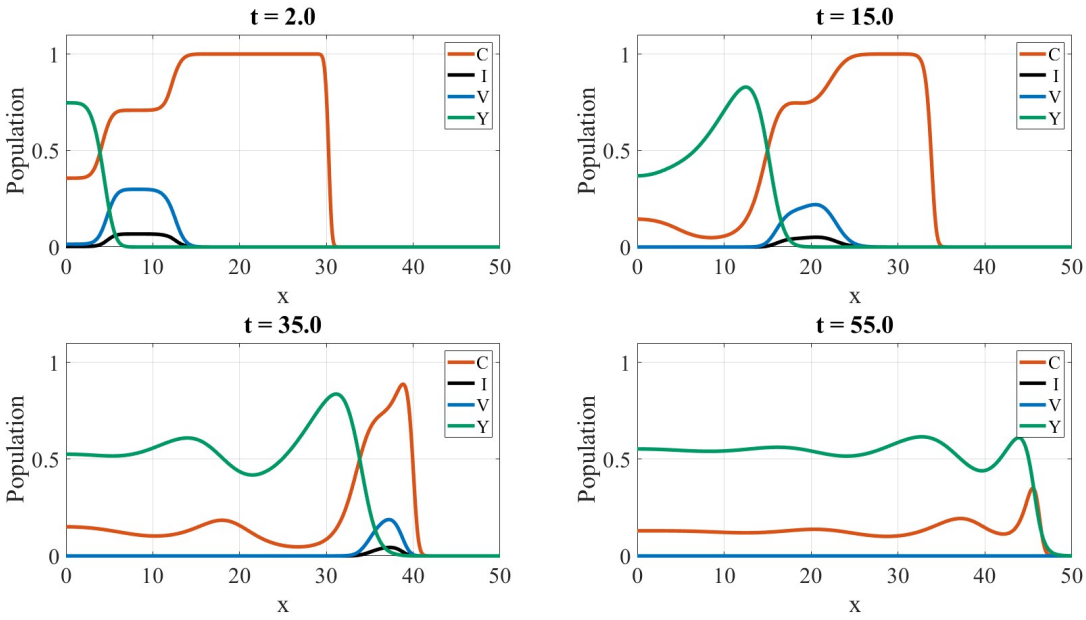


Figure 9: The traveling waves of the four populations invading to the right. The waves are captured at $t = 2,\ 15,\ 35$, and 55.

by a factor of ten, the immune invasion speed in (55) becomes

$$c_a^* = 2\sqrt{0.25(\eta C_2 + \zeta I_2 - \xi)} \approx 0.961, \tag{63}$$

which is faster than the virus invasion speed.

Simulations for this parameter regime are shown in Figure 9. The immune front catches up with the viral infection front, clears the virus population, and subsequently reaches the boundary of the cancer region. Between $t = 15$ and $t = 35$, the immune front advances by $\Delta X = 20.658$, yielding a numerical invasion speed of $c^* \approx 1.032$. This value is consistent with the theoretical prediction $c^* \approx 1.058$ obtained in (63). Since only the immune-cell diffusion coefficient is modified, the invasion speeds of the remaining populations are unchanged.

Case	Wave	Theoretical	Numerical
Slow immune	cancer invasion	$c_c^* = 0.316$	$c_{c_n}^* = 0.312$
	virus invasion into the cancer	$c_v^* = 0.930$	$c_{v_n}^* = 0.900$
	immune invasion into cancer	$c_y^* = 0.335$	$c_{y_n}^* = 0.335$
	immune invasion into virus	$c_a^* = 0.304$	$c_{a_n}^* = 0.304$
Fast immune	cancer invasion	$c_c^* = 0.316$	$c_{c_n}^* = 0.312$
	virus invasion into the cancer	$c_v^* = 0.930$	$c_{v_n}^* = 0.900$
	immune invasion into cancer	$c_y^* = 1.058$	$c_{y_n}^* = 1.032$
	immune invasion into virus	$c_a^* = 0.961$	$c_{a_n}^* = 0.973$

Table 4: Comparison of theoretical and numerical wave speeds for slow versus fast immune response.

5 Conclusion

The use of oncolytic viruses as a cancer treatment has attracted considerable attention. In fact, some of these viruses have already passed clinical trials and have received FDA approval for clinical use (Greig, 2016). However, oncolytic virotherpay has not reached its maximum efficacy (Russell and Barber, 2018) due to physical barriers, tumor heterogeneity, and an immunosuppressive tumor microenvironment. To further investigate the role of the immune response for OVT, we developed a reaction-diffusion model, inspired by the models by Baabdulla and Hillen (2024) and Al-Tuwairqi et al. (2020). Our model was designed to capture the complex dynamics between cancer cells, oncolytic viruses, and the immune system. In this work, we conduct a detailed qualitative and quantitative analysis of the system in order to address the question: why, despite their promise, do oncolytic viruses as a monotherapy rarely achieve complete and lasting regression of established tumors?

By incorporating an immune cell population into the model of Baabdulla and Hillen (2024), we have taken a step toward making the OVT model more biologically realistic. However, several limitations remain. A more comprehensive representation of the immune response would require distinguishing between innate and adaptive immune components, and further subdividing the adaptive response into tumor-specific and virus-specific populations (Storey et al., 2020). Despite these potential extensions, the model still cannot fully capture the complex interactions among the tumor, virus, and immune system. Increasing biological detail introduces additional equations and parameters, which significantly complicates the mathematical analysis. Therefore, to maintain analytical tractability while retaining the essential immune effects, we represent the immune system by a single additional variable Y, to represent the overall effect of the immune response

The dynamics of the basic oncolytic virus model have been extensively studied. In particular, a Hopf bifurcation occurring at sufficiently large burst sizes gives rise to periodic solutions. These oscillatory dynamics are characterized by repeated reductions in the cancer cell population to low levels between phases of regrowth, which, according to the model, may increase the likelihood of tumor eradication. The inclusion of the immune response significantly alters this behavior. For very weak immune responses, the parameter range supporting periodic oscillations is largely preserved. However, as the immune response strengthens, the periodic regime disappears. A strong immune response not only targets tumor cells but also attacks the virus, potentially clearing it before it can infect the entire tumor population. This behavior is reflected in the bifurcation diagram in Figure 2, where a large **green** region indicates parameter values for which oscillations are suppressed.

For large values of θ, we observe the opposite effect: the immune system becomes effectively suppressed, leading to immune-free dynamics. Our analysis suggests that the combination of oncolytic virus and immune response is favorable for treatment within the model for intermediate values of η and θ, where a balance between viral replication and immune activity is achieved. This influence of the immune response is further supported by the stacked wave analysis. In Figure 9, the immune wave catches up with the viral infection front and eliminates it. This behavior is robust within our model and may represent a key mechanism underlying OVT failure. The sensitivity analysis reinforces this conclusion, showing that treatment outcomes are most sensitive to the immune stimulation parameter

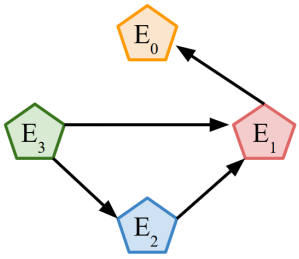


Figure 10: Schematic of the heteroclinic connections between equilibria that comprise the stacked wave. Each of these connections represents an invasion front that might travel at different speeds.

η, the immune exhaustion rate ξ, and the viral burst size θ. Therefore, successful treatment requires a delicate balance between viral infectivity and immune efficiency. Our analytical curve $\theta_1(\eta)$ derived in (41) provides an explicit threshold relation: for $\theta > \theta_1(\eta)$, the model predicts successful treatment. These theoretical predictions offer testable hypotheses for experimental validation.

The stacked waves found here are also raises several mathematical questions. Similar to the Fisher-KPP equation, travelling waves can be related to heteroclinic orbits in phase space (de Vries et al., 2006). A travelling wave analysis of our model (44) would lead to an 8 dimensional phase space. Numerically, we find heteroclinic connections from E_1 to E_0, from E_2 to E_1, from E_3 to E_1, and from E_3 to E_2, **albeit for different speeds**. These connections **can be illustrated as a connected** tree as in Figure 10. To prove the existence of such a connected tree of heteroclinic orbits is a formidable challenge, which is beyond the scope of the present work.

The identification of stacked travelling waves has important medical implications. We distinguish four invasion speeds: c_c for cancer invasion into healthy tissue, c_v for viral invasion into tumor tissue, c_y for immune cell invasion into uninfected tumor tissue, and c_a for immune cell invasion into virus-infected tumor tissue. The relative ordering of these speeds depends on the model parameters and ultimately determines the clinical outcome.

For example, in Figure 8 and Table 4, we observe the ordering $c_a < c_c < c_y < c_v$. In this case, the model predicts that the immune response alone is capable of catching up with the cancer invasion front. However, when combined with OVT, the immune wave is slowed down, allowing the cancer to outrun the immune response. At the same time, the virus is able to catch up with the advancing tumor front.

In contrast, for a faster immune response (Figure 9 and Table 4), the ordering becomes $c_c < c_v < c_a < c_y$. For this parameter set, all invasion fronts eventually catch up with the tumor front. Notably, the immune invasion speed into virus-infected tumor tissue is lower than the corresponding invasion speed into uninfected tumor tissue. However, this ordering is not universal. The explicit formulas for c_a^* (61) and c_y^* (55), show that the immune invasion speed is enhanced in the presence of the virus provided that

$$\eta C_2 + \zeta I_2 - \xi > \eta - \xi,$$

which simplifies to the condition

$$\zeta > \eta \frac{\theta(\gamma + \theta)}{\gamma}.$$

For the parameter values listed in Table 3, this inequality requires $\zeta > 6.8$, corresponding to an immune stimulation rate that is substantially larger than the baseline value. Consequently, for the parameter regimes considered here, immune invasion into virus-infected tumor tissue is predicted to be slower than immune invasion into uninfected tumor tissue.

Several mechanisms underlying OVT failure have been proposed in the literature.The present model suggests an additional mechanism:: treatment failure may occur when viral spread is insufficiently rapid to outpace both tumor invasion and the immune response.

Our travelling wave analysis is performed on the nondimensional system (3). To facilitate biological interpretation, we now convert the corresponding wave speeds back to dimensional form. From the nondimensionalization given in (2), we obtain

$$\frac{dx}{dt} = \sqrt{rD_V}\,\frac{d\hat{x}}{d\hat{t}}, \tag{64}$$

which implies that the dimensional wave speeds satisfy

$$c = \sqrt{rD_V}\,c^*.$$

The resulting dimensional invasion speeds are reported in Table 5.

Case	Wave	Theoretical
Slow immune	cancer invasion	$c_c = 0.0134$ mm/day
	virus invasion into the cancer	$c_v = 0.0349$ mm/day
	immune invasion into cancer	$c_y = 0.0142$ mm/day
	immune invasion into virus	$c_a = 0.0128$ mm/day
Fast immune	cancer invasion	$c_c = 0.0134$ mm/day
	virus invasion into the cancer	$c_v = 0.0349$ mm/day
	immune invasion into cancer	$c_y = 0.0448$ mm/day
	immune invasion into virus	$c_a = 0.0407$ mm/day

Table 5: Waves propagation speeds in dimensional coordinates.

Many additional combinations of invasion speeds are possible. Explicit formulas for several of these speeds are given in (46), (55), and (61), where the wave speeds are expressed directly in terms of the model parameters. Once these parameters are specified, quantitative estimates of the corresponding invasion speeds can be obtained.

The stacked-wave analysis further suggests that the relative timing of the invasion fronts influences the predicted treatment outcome. If the viral front propagates substantially faster than the immune front, immune-mediated viral clearance may occur only after the virus has begun to decline, potentially allowing tumor regrowth. Conversely, if the immune wave propagates too rapidly, it may wipe out the virus prematurely, which is a common problem in clinical trials of oncolytic virotherapy. **These observations indicate that the timing of virotherapy may influence treatment efficacy. In particular, treatment schedules that improve the overlap between viral spread and the immune response may enhance**

therapeutic outcomes. While no clinical paper explicitly says "stacked waves," the phenomenon of multi-phase tumor response (rapid shrinkage followed by slower stabilization or regrowth) are qualitatively consistent with the dynamics predicted by the model. In the future, stacked-wave analysis might guide design of combination therapies, optimizing virus dosing schedules that balance the speeds of viral spread and manipulating immune response for maximum tumor eradication.

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Data Availability. No data were generated or analyzed during the course of this study, so data sharing is not applicable.

Conflict of Interest. The authors declare that they have no competing interests.

A Base parameters

Here, we briefly explain the choice of model parameters as listed in Table 2 and their ranges. The tumor diffusion rate, D_C, was taken from Baabdulla and Hillen (2024), who adopted it from Pooladvand et al. (2021). Similarly, the virus diffusion rate, D_V was taken from the same sources, and in Baabdulla et al. (2025) it was further fitted and validated against experimental data for reovirus spread on breast cancer cultures. We assumed that infected tumor cells diffuse at the same rate as uninfected tumor cells. Moreover, for simplicity, we assumed that immune cells diffuse at the same rate (D_Y) as tumor cells.

The infection rate β was chosen to be the same as in Baabdulla and Hillen (2024); Pooladvand et al. (2021). In the parameter fitting performed by Baabdulla et al. (2025), the death rate of infected tumor cells a, the number of viruses released per lysed infected tumor cell b, and the virus-induced immune response rate ω were fitted and validated against experimental data for the oncolytic virus. We adopted the same parameter values. In the experiments carried out by Kim et al. (2006) on adenovirus in the glioblastoma U343 cell line, the tumor growth rate r was estimated to be approximately 0.3 per day. This value was used by Baabdulla and Hillen (2024); Pooladvand et al. (2021) and also lies within the range reported by Storey et al. (2020), and we will use this value in our analysis.

With slight modifications, the immune-related parameters—accounting for natural killer cell activity and cytokine-mediated effects—were adopted from Al-Tuwairqi et al. (2020), who in turn based their values on Friedman et al. (2006). The latter study investigated an OVT model using a mutant herpes simplex virus type 1 (hrR3) for glioma treatment.

Specifically, the killing rate of uninfected tumor cells by immune cells (p_1), the killing rate of infected

tumor cells by immune cells (q_1), the killing rate of virus by immune cells (s_1), the stimulation rate of immune cells by uninfected tumor cells (η_1), and the stimulation rate of immune cells by infected tumor cells (ζ_1) were taken from these sources. In our simulations, we retained the same parameter values for p_1 and ζ_1. Since infected tumor cells are expected to be more susceptible to immune attack, we chose $q_1 > p_1$ and adjusted its value accordingly. To avoid overly rapid viral clearance by the immune system, we set s_1 to approximately one half of p_1.

Furthermore, because immune stimulation by infected cells differs from that by uninfected cells, we halved the value of ζ_1 and adopted it for the value for η_1. Finally, the immune-cell clearance rate was set to 0.036, which lies within the range reported by Storey et al. (2020).

B Inclusion of virus induced immune stimulation

Inspired by the work of Al-Tuwairqi et al. (2020), in the base model (4), we did not include a direct stimulation of the immune system by free virus. In this section, we study how the inclusion of this mechanism alters the results obtained in the previous analysis. We model this effect by introducing a mass-action term ρYV in the immune equation, describing direct immune activation through interactions with the virus. The last equation of model (3) that describes the dynamics of the immune population Y then becomes

$$\frac{\partial Y}{\partial t} = D_Y \Delta Y + \eta YC + \zeta YI + \rho YV - \xi Y. \tag{65}$$

We assume that immune stimulation by free virus is weaker than stimulation by infected cancer cells and we assume ρ is small compared to ζ. Specifically we use a small parameter $0 < \delta < 1$ and consider $\rho = \delta\zeta$.

The additional term in the Y equation does not change the overall bifurcation structure as shown in Figure 11, but it slightly alters the parameter values where transitions between stable, unstable, and periodic arise. It also affects the values of the wave–front speeds that we calculated above.

Of particular interest is the case where the virus infection can actually speed up the immune invasion wave. Following calculations analogous to those leading to (61), the minimal wave speed of immune invasion into the virus is now given by

$$c_\delta = 2\sqrt{D_y(\eta C_2 + \zeta I_2 + \rho V_2 - \xi)}. \tag{66}$$

This speed is certainly larger than c_a from (61) due to the additional term.

We again compare the immune–virus invasion speed to the immune–cancer invasion speed, and consider the case when

$$c_\delta > c_y.$$

Substituting $\rho = \delta\zeta$ and simplifying yields the condition

$$\delta > \frac{\eta}{\zeta\theta}(\gamma + \theta) - \frac{\gamma}{\theta}. \tag{67}$$

Using the parameter values in Table 3, this inequality is satisfied for $\delta > 0.375$. We therefore choose

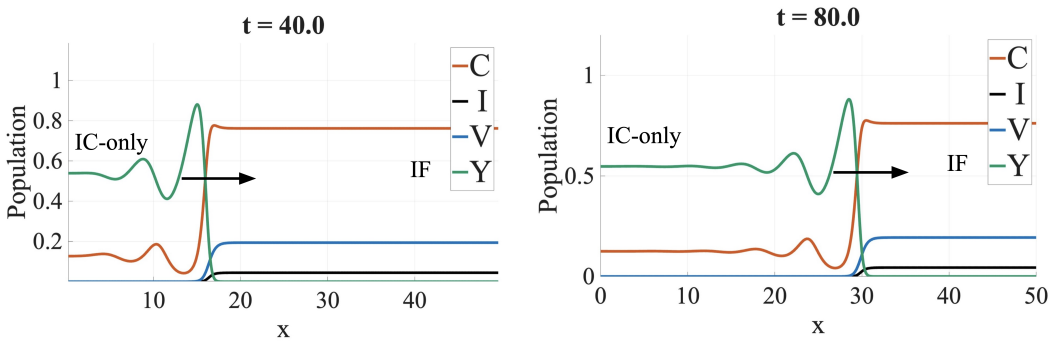


Figure 11: The immune-cancer only (IC-only) invading immune-free(IF) equilibria. The wave moved from $X_1 = 15.928$ in $t = 40$ to $X_2 = 29.415$ at $t = 80$.

$\delta = 0.4$, corresponding to $\rho = 0.4\,\zeta$, and present the resulting dynamics in Figure 11. We obtain the invasion speed through the equation (66) $c_\delta = 0.337$ which matches the numerical value estimation of 0.337. This value is greater our previously calculated value of $c_y = 0.335$ for immune invasion into cancer (see Table 4). Hence here the scenario where the viral infection speeds up the immune wave is a relevant case.

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